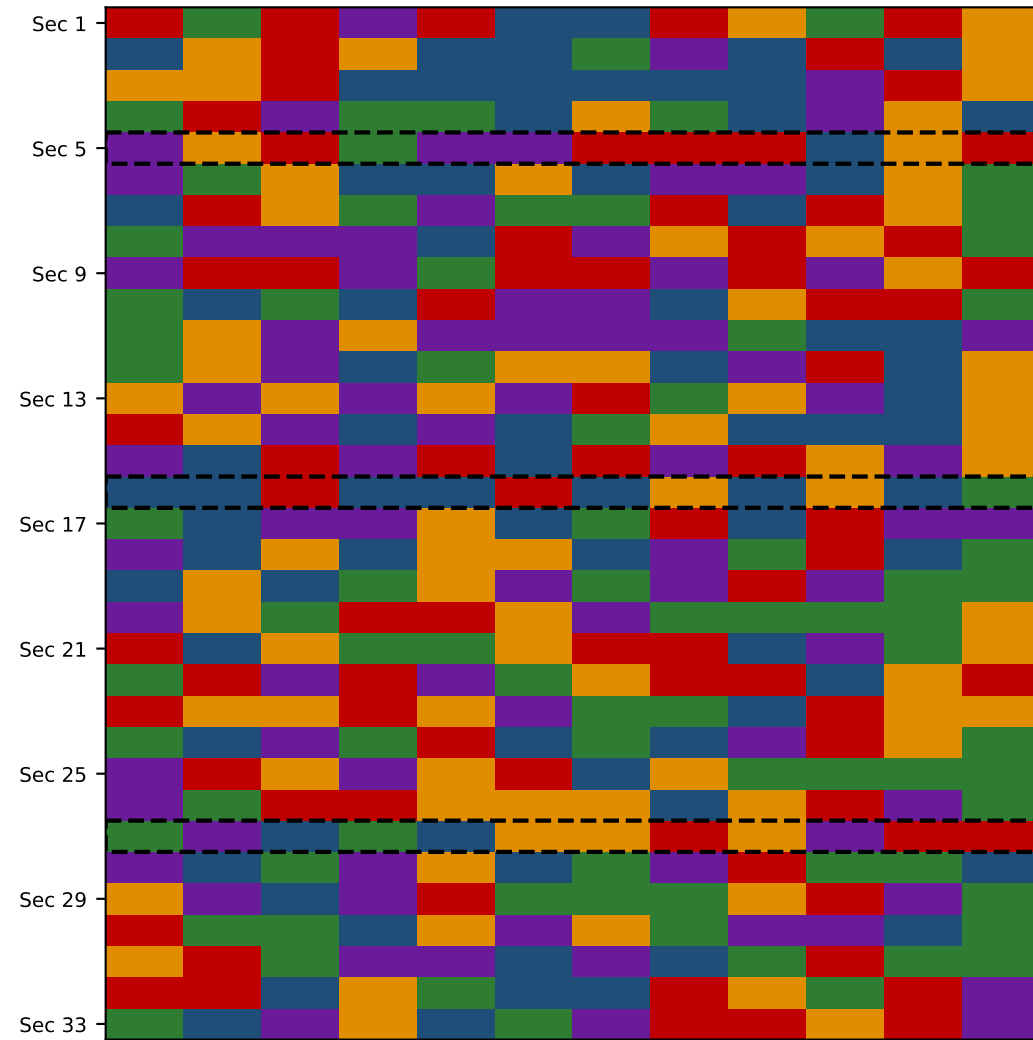


Why an ungrouped split inflates reported accuracy on section-structured pavement data (33 sections × ~12 observations, matching the CRCP substrate audited in Section 7)

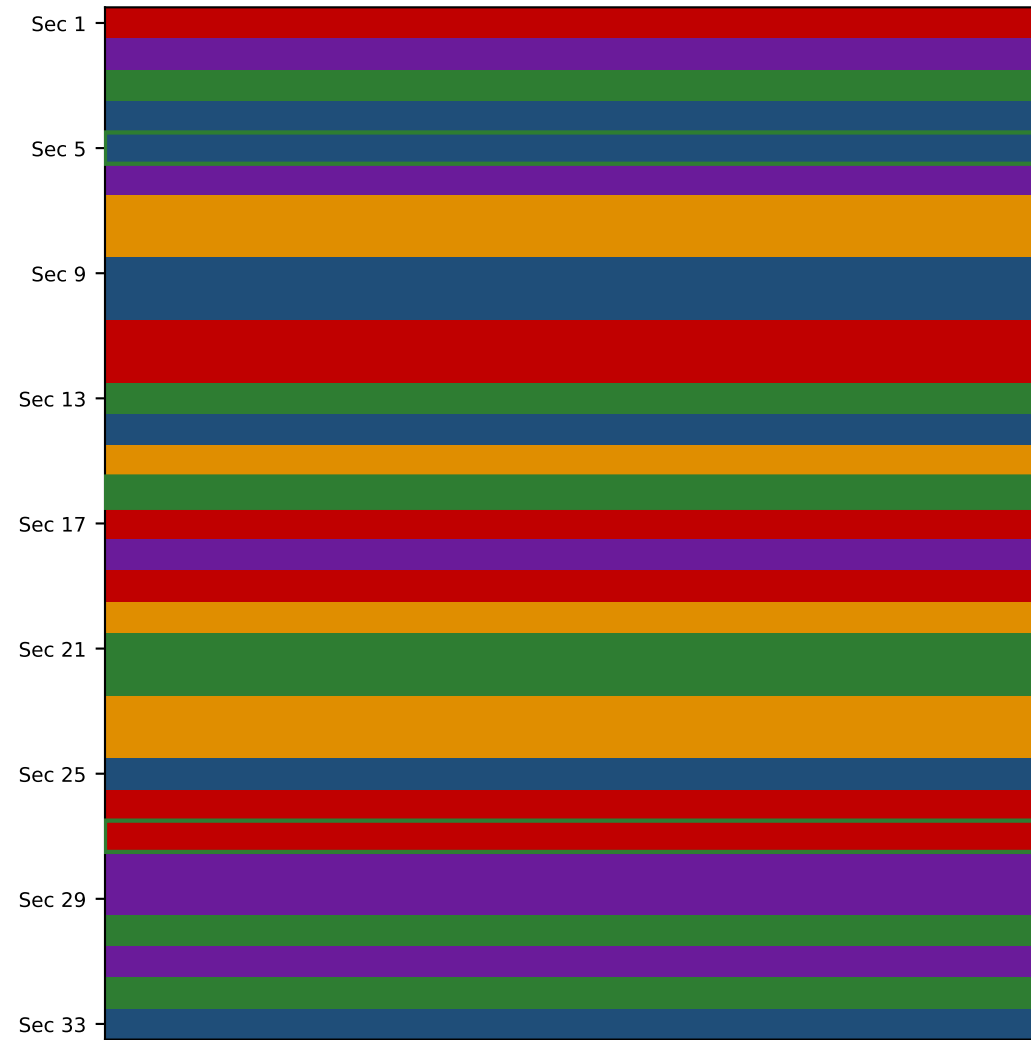
In Panel A, a model can see 10-11 of a section's 12 observations in training and be "tested" on the 12th — near-duplicate, spatially correlated records leak across the train/test boundary. In Panel B, an entire section is held out together, so the test fold contains no information the model could have seen during training.

A. Random 5-fold split (observation level)



As reported: "the dataset was split into 5 subsets..."
(Alnaqbi et al., full-text confirmed) — dashed boxes: one section, multiple folds

B. Section-grouped 5-fold split



Alternative not used in the reviewed papers
solid boxes: one section stays entirely within one fold

Fold 1 Fold 2 Fold 3 Fold 4 Fold 5